Two-Part e-Unimodality of the Chromatic Quasisymmetric Functions of Triangular Ladders

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Abstract

Let $G_n = P_n^2$ be the square of the path on n vertices, also called a triangular ladder. We establish that every coefficient indexed by a two-part partition in the elementary-symmetric-function expansion of its chromatic quasisymmetric function is a nonnegative, palindromic, unimodal polynomial in q. Specifically, for $n \geq 3$ and $1 \leq k \leq \lfloor n/2 \rfloor$, the coefficient $[e_{n-k,k}]X_{G_n}(\mathbf{x}; q)$ has these properties. The proof uses the decreasing-tree expansion in Tom's Proposition 5.3 and Lemma 3.13, a four-state transfer matrix, coefficient extraction, and positive symmetric-unimodal decompositions. This statement is restricted to two-part coefficients and does not prove full e-unimodality. Mathematical checks were conducted computationally and subjected to an independent AI-assisted adversarial audit; no human peer review has been conducted. A literature search did not find an identical theorem among the works inspected; novelty remains inconclusive.

Keywords. Chromatic quasisymmetric function; triangular ladder; elementary symmetric function; unimodality; transfer matrix; decreasing tree.
Classification. Research preprint; independent, AI-assisted work.
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Verification status. Proof reviewed by an independent AI system, conditional on the cited primary results; not human peer reviewed. Novelty: INCONCLUSIVE.

1 Theorem and conventions

For $n \geq 3$, let $G_n = P_n^2$ have vertex set $[n] = \{1, \ldots, n\}$ and edges $\{i, j\}$ whenever $1 \leq |i-j| \leq 2$. This is $G_{n,3}$ in the Shareshian–Wachs notation and is also called $P_{n,2}$, the triangular ladder. We use the chromatic quasisymmetric function $X_{G_n}(\mathbf{x}; q)$, with ascents corresponding to edges $\{i, j\}$ with $i < j$ and color $\kappa(i) < \kappa(j)$. Write $[r]_q = 1 + q + \cdots + q^{r-1}$ for $r \geq 1$ and $[0]_q = 0$; put $t = 1 + q$.

Theorem 1 (Two-part e-unimodality)**.** *For every $n \geq 3$ and every $1 \leq k \leq \lfloor n/2 \rfloor$, the polynomial*

$$c_{n-k,k}(q) := [e_{n-k,k}]X_{G_n}(\mathbf{x}; q)$$

has nonnegative integer coefficients, is palindromic, and is unimodal in q. For $(n, k) = (3, 1)$ and $(4, 2)$ this coefficient is zero (and the assertion is interpreted trivially).

Only two-part partitions are considered; no statement is made here for partitions with three or more parts.

2 Imported decreasing-tree identities

We use Proposition 5.3, equations (5.3)–(5.5), and Lemma 3.13 of Tom [1]. In particular,

$$s_j(G) = \sum_{(T_1,T_2)} q^{\mathrm{inv}_G(\mathrm{list}(T_1)\mathrm{list}(T_2))}, \tag{1}$$

where the ordered pair of decreasing trees partitions $V(G)$, vertex 1 belongs to T_1, and $|V(T_2)| = j$. The weight in (1) has *no* additional $\sum_i (r_i - 1)$ term; the general forest-triple weights elsewhere in Tom's paper are distinct from this specialized definition.

For G_n, define $s(G_n) = \prod_{i=2}^{n} [b_i]_q = t^{n-2}$ (with b_i the number of smaller neighbors of i). Define

$$U_{n,j}(q) = [j]_q s_j(G_n) - t^{n-2}. \tag{2}$$

For $1 \le k < n/2$, Tom's expression becomes

$$c_{n-k,k}(q) = [n-k]_q U_{n,k}(q) + [k]_q U_{n,n-k}(q). \tag{3}$$

For $n = 2k$, the equal-part exception uses *one* term only:

$$c_{k,k}(q) = [k]_q U_{2k,k}(q). \tag{4}$$

These cited results are used as established inputs; their proofs are not reproduced here.

3 Four-state transfer matrix and generating function

Let $A = V(T_1)$, $B = V(T_2)$, so $1 \in A$. For a fixed subset S, the decreasing-tree weight sum is, by Tom's Lemma 3.13,

$$T(S) = \prod_{i \in S,\ i > \min S} [b_i(S)]_q, \qquad b_i(S) = \#\{j \in S : i - 2 \le j < i\}.$$

The count $b_i(S)$ is taken *inside* S. The cross-block inversion exponent equals

$$\mathrm{cross}(A,B) = \#\{(b,a) : b \in B,\ a \in A,\ b < a \le b + 2\}.$$

We enumerate assignments of the vertices in increasing order, with z marking the number of B vertices and x marking total vertices. Prior to the first B there is a nonempty run of A's; the first B occurs at position $j \ge 2$. Its initial generating weight is

$$I(x,z) = zx^2 + \sum_{j \ge 3} zx^j t^{j-3} = \frac{zx^2(1-qx)}{1-tx}.$$

Once the first B appears, keeping only the two most recent assignments suffices. In order AA, AB, BA, BB, appending a vertex has transition matrix

$$M = \begin{pmatrix} t & 0 & 0 & 0 \\ 0 & 0 & q & z \\ q & z & 0 & 0 \\ 0 & 0 & 0 & zt \end{pmatrix}.$$

For example, appending an A requires a previous A in the last two positions (otherwise the new vertex cannot enter a decreasing tree on A), contributes $[\#\text{ preceding } A]_q$, and contributes an inversion factor $q^{\#\text{ preceding } B}$ within that two-position window. Appending a B

analogously contributes $z[\#\text{ preceding } B]_q$. This yields the displayed transitions, including zeros for forbidden moves.

Let e_{AB} be the coordinate vector of AB, and $\mathbf{1}$ the all-ones column vector. Then the bivariate generating function from (1) is

$$\begin{aligned}\mathcal{S}(x,z;q) &= \sum_{n\geq 2}\sum_{1\leq j<n} s_j(G_n)x^n z^j \qquad (5)\\ &= I(x,z)e_{AB}^{\mathsf{T}}(\mathrm{Id}-xM)^{-1}\mathbf{1}\\ &= \frac{zx^2(1-qx)}{(1-tx)(1-qzx^2)}\left(\frac{1-x-qx^2}{1-tx}+\frac{xz}{1-tzx}\right). \qquad (6)\end{aligned}$$

The last equality is a rational identity; the combinatorial interpretation follows from the local weights and Tom's decreasing-tree lemma above.

4 Explicit coefficient extraction

Fix $n = 2k + d$, where $k \geq 1$ and $d \geq 0$. For $k \geq 2$ set

$$C_k(q) = \sum_{a=0}^{k-2} q^a t^{2k-3-2a}, \qquad C_1(q) = 0. \qquad (7)$$

Using $(1-qx)(1-x-qx^2) = 1-tx+q^2x^3$, split (6) into

$$\mathcal{S}_I = \frac{zx^2(1-tx+q^2x^3)}{(1-tx)^2(1-qzx^2)},$$
$$\mathcal{S}_{II} = \frac{z^2x^3(1-qx)}{(1-tx)(1-tzx)(1-qzx^2)}.$$

The needed identities, with $N \geq 0$, are

$$[x^N]\frac{1-tx+q^2x^3}{(1-tx)^2} = t^N + \mathbf{1}_{N\geq 3}(N-2)q^2t^{N-3}, \qquad (8)$$

$$[x^J]\frac{1-qx}{1-tx} = \begin{cases}1 & J=0,\\ t^{J-1} & J\geq 1,\end{cases} \qquad (9)$$

where the second identity uses $t - q = 1$.

For $s_k(G_{2k+d})$, extraction of z^k from $\mathcal{S}_I$ forces $q^{k-1}x^{2k}$, yielding $q^{k-1}t^d + \mathbf{1}_{d\geq 3}(d-2)q^{k+1}t^{d-3}$. In $\mathcal{S}_{II}$, expansion indices r, s satisfy $r+s = k-2$; the remaining x-degree is $J = k+d-1-r \geq 1$. Formula (9) then gives $\sum_{r=0}^{k-2} q^r t^{n-4-2r}$. Hence

$$\boxed{s_k(G_n) = q^{k-1}t^d + \sum_{r=0}^{k-2} q^r t^{n-4-2r} + \mathbf{1}_{d\geq 3}(d-2)q^{k+1}t^{d-3}.} \qquad (10)$$

For $d = 0$ the displayed sum is used directly, with no negative exponents introduced.

For $s_{n-k}(G_n)$ when $d \geq 1$, the $\mathcal{S}_I$ portion cannot contribute, because its least x-degree for z^{n-k} is $2(n-k) > n$. The $\mathcal{S}_{II}$ index r lies in $0 \leq r \leq k-1$, with remaining degree $J = k-1-r$. The terms $r \leq k-2$ yield the same interior sum; the boundary $r = k-1$ has $J = 0$ and contributes $q^{k-1}t^{d-1}$. Therefore

$$\boxed{s_{n-k}(G_n) = t^{d-1}\big(q^{k-1} + C_k(q)\big), \quad d \geq 1.} \qquad (11)$$

Finally, the finite geometric series in (7) gives, for $k \geq 2$,

$$\boxed{[3]_q C_k = t^{2k-1} - q^{k-1}t,} \tag{12}$$

using $t^2 - q = [3]_q$.

5 Diagonal D_k and the smaller-index terms

Set $D_k(q) = U_{2k,k}(q)$. Substituting $d = 0$ in (10) yields $D_1 = D_2 = 0$: for $k = 1$ the sum is empty; for $k = 2$, $C_2 = t$, and $[2]_q(q+1) - t^2 = 0$. For $k \geq 3$ combine (10) with (12):

$$[3]_q D_k = q^3[k-3]_q t^{2k-2} + q^k t[k]_q. \tag{13}$$

The division by $[3]_q$ is exact because D_k is a polynomial by definition. Sum (13) over $k \geq 3$ using geometric series to obtain

$$\boxed{\sum_{k\geq 3} D_k(q) y^{k-3} = \frac{q^3 t(1+qty)}{(1-qy)(1-q^2y)(1-t^2y)(1-qt^2y)}.} \tag{14}$$

Its coefficient extraction is a *positive decomposition*, with empty sums interpreted as zero:

$$\begin{aligned} D_k = q^3 t \sum_{a+b=k-3} q^a [a+1]_q t^{2b} [b+1]_q \\ + q^4 t^2 \sum_{a+b=k-4} q^a [a+1]_q t^{2b} [b+1]_q. \end{aligned} \tag{15}$$

Every nonzero summand has nonnegative coefficients and is symmetric and unimodal around the same center $(3k-2)/2$. Products of symmetric unimodal polynomials with nonnegative coefficients remain symmetric unimodal; aligned-center sums preserve this property. Thus all D_k have the property.

Subtracting t^{n-2} in (2) and comparing (10) at d and 0 gives

$$\boxed{U_{n,k} = t^d D_k + \mathbf{1}_{d\geq 3}(d-2) q^{k+1}[k]_q t^{d-3}.} \tag{16}$$

Both nonzero pieces of (16) are symmetric unimodal with common center $(n+k-2)/2$. Hence $U_{n,k}$ has those properties for all $1 \leq k \leq n/2$.

6 The larger-index terms

For $d \geq 1$, formula (11) yields

$$U_{n,n-k} = t^{d-1}\left([k+d]_q\left(q^{k-1} + C_k\right) - t^{2k-1}\right).$$

For $k \geq 2$, use (12) and $[k+d]_q = [3]_q + q^3[k+d-3]_q$. The intermediate steps are

$$\begin{aligned} [3]_q(q^{k-1} + C_k) - t^{2k-1} &= q^{k-1}([3]_q - t) = q^{k+1}, \\ q^3[k+d-3]_q(q^{k-1} + C_k) &= q^3[k+d-3]_q C_k + q^{k+2}[k+d-3]_q, \\ [k+d]_q(q^{k-1} + C_k) - t^{2k-1} &= q^3[k+d-3]_q C_k + q^{k+1}(1 + q[k+d-3]_q). \end{aligned}$$

With $1 + q[m]_q = [m+1]_q$ we obtain

$$\boxed{U_{n,n-k} = t^{d-1}\left(q^3[k+d-3]_q C_k + q^{k+1}[k+d-2]_q\right), \quad k \geq 2.} \tag{17}$$

The case $k = 1$ is separate:

$$U_{n,n-1} = q^2[d-1]_q t^{d-1}, \qquad n = 2+d,\ d \geq 1. \tag{18}$$

Each summand in C_k has center $(2k-3)/2$. For (17), the two nonzero terms before multiplication by t^{d-1} have common center $(3k+d-1)/2$; afterwards it becomes $(2n-k-2)/2$. The same conclusion applies to (18). Thus $U_{n,n-k}$ is nonnegative, symmetric, and unimodal for all $n > 2k$.

7 Proof of Theorem 1

When $n > 2k$, both terms on the right of Tom's (3) are nonnegative symmetric unimodal by (16) and (17)–(18). Their centers coincide:

$$\underbrace{\frac{n-k-1}{2} + \frac{n+k-2}{2}}_{[n-k]_q U_{n,k}} = \underbrace{\frac{k-1}{2} + \frac{2n-k-2}{2}}_{[k]_q U_{n,n-k}} = \frac{2n-3}{2}.$$

Their sum is therefore also symmetric unimodal, with nonnegative coefficients. When $n = 2k$, Tom's single-term identity (4) and the decomposition for D_k give the same conclusion. The two zero cases stated in the theorem are immediate. This proves Theorem 1 conditional on the established cited identities of Tom [1]. □

8 Prior work, scope, and reproducibility

The full e-unimodality problem for triangular ladders, involving all partition shapes, is distinct from the two-part statement treated here; see Tom [1]. Cho–Huh [2, 3] prove e-unimodality for special natural unit interval graph families. Their publicly inspected 2018 theorem hypotheses do not contain P_n^2 for all $n \geq 6$; however, the full theorem list of the 2019 SIAM journal edition has *not* been checked line by line. Sagan–Tom [5] and Dahlberg [4] address related questions but the examined statements did not give an identical all-n two-part q-unimodality theorem. This is not proof that no equivalent result exists; the novelty assessment is **INCONCLUSIVE**.

The associated reproduction archive includes the v0.4 audit proof, a four-state rational identity check, finite coefficient checks, direct-coloring basis-change checks, and SHA-256 manifests. Finite calculations test the formulas but do not prove the universal statement. The authoring process was AI assisted, with an independent AI-assisted adversarial audit reported on October 31, 2026; this paper has **not** undergone human peer review. Claimed scope is deliberately confined to two-part partitions.

References

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